

Synthetic Fair-Value Strategy

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1. Introduction

A prediction-market contract is a binary claim that pays \$1.00 if an outcome occurs and \$0.00 if it does not; before settlement it trades between those bounds at whatever price buyers and sellers agree reflects the outcome's probability. Polymarket US launched regulated, real-money contracts of this kind in January 2026, and its coverage of the 2026 FIFA World Cup listed, for each tracked player in each match, three separate contracts on a central limit order book: at least one goal, at least one assist, and at least one goal-or-assist. Each has its own order book and its own update cadence.

These three contracts are not independent instruments. Scoring a goal is one of the ways to record a goal-or-assist, so the compound contract can never be worth less than either single-stat leg:

$$P(\text{goal} \cup \text{assist}) \geq \max(P(\text{goal}), P(\text{assist})).$$

When the compound contract is quoted less actively than the legs that decompose it, its price drifts below that floor. This is a pricing error the market makes against itself, and it can be measured without taking a position. Across 157,595 simultaneously captured three-leg quotes, the compound bid falls below the goals bid in 28.81% of pre-kickoff observations, and the compound contract carries a 3.81× wider spread and rests 5.6× less size than the goals leg it can be priced against.

Whether that staleness reverts is a separate question, and it is tested rather than assumed. Conditioning on positions entered while the compound contract was cheap relative to its formula fair value, the mispricing narrows toward fair value 71.4% of the time, against 8.5% under a volatility-matched random walk ($z = 8.43$, $p < 0.000001$). A second strategy trades confirmed public events directly: when a goal, a substitution, or a final whistle makes one side of a contract near-certain, a resting order on the other side has not yet requoted, and whoever acts before the book adjusts is trading a latency gap.

The purpose of this paper is to measure the pricing inefficiency, test whether it is a consistent and statistically significant effect, and evaluate whether either strategy built around it beats the market's own pricing rather than merely trading faster than it. Only the patience strategy does. The speed strategy is profitable and never lost a position it held to settlement, but because it buys contracts already priced near certainty, its win rate is the arithmetic of those prices and not a measurable edge, a distinction Section 5 makes precise.

2. Data Pipeline and Screening Architecture

This analysis relies on a fully automated data pipeline that identifies, validates, and tracks pre-kickoff prop markets while minimizing contamination from fundamentally driven repricing. The objective is a clean sample of simultaneously observed three-leg quotes and a measurement of price behavior in the window where the compound contract de-stales.

The pipeline runs unattended on Windows Task Scheduler and integrates three data sources. First, market and event data is read from Polymarket US's public gateway API, which carries a live score and match clock on each event object. Second, the account's own fills, balances, and settlements are read from the authenticated portfolio API, signed with an Ed25519 key over the message `{timestamp}{method}{path}`. Third, event timestamps used for latency measurement are drawn from neutral match reports (FIFA, ESPN, Al Jazeera), never from Polymarket's own price data, so that the latency figures are not circular.

Polling is tiered by time to kickoff: every twenty minutes beyond six hours out, tightening to every sixty seconds inside the final thirty minutes. Market discovery is structural rather than a hand-maintained list. The set of soccer leagues is pulled live from the platform's own sports endpoint, which returned seventy leagues besides the World Cup as of 2026-08-08, and each league's games are then enumerated against the API.

A clean three-leg observation is recorded only when the goals, assists, and goals-or-assists contracts for the same player can all be read from the same market state in a single pass. Pairing all three legs at identical capture timestamps yields 157,595 simultaneous observations, of which 124,571 fall in the forty-eight hours before kickoff. This is the sample used throughout the paper.

Every fill the account made is reconciled at ingestion: each trade row's quantity is checked against its cost divided by its price, and the local trade log is cross-checked one-to-one by trade identifier against a fully paginated pull of the account's complete live history (392 trades). The trading client used for these reads exposes only balance, position, and activity lookups, and contains no order-placement methods.

3. Signal Identification

For a given player in a given match, let g , a , and ga be the traded prices of the “1+ goals,” “1+ assists,” and “1+ goals-or-assists” contracts. Under a statistical-independence approximation, inclusion–exclusion gives a fair value for the compound contract:

$$P(\text{goal} \cup \text{assist}) = P(\text{goal}) + P(\text{assist}) - P(\text{goal} \cap \text{assist}) \approx g + a - g a.$$

The signal is the difference between that fair value and the contract's actual price:

$$\text{gap} = (g + a - g a) - ga.$$

A positive gap means the compound contract is trading cheap relative to its own legs, which is a buy signal. Across 208 clean pre-kickoff samples, gap averaged +0.0106 and exceeded 0.05 in absolute value in 31 of 208 cases. On an instrument priced between \$0.00 and \$1.00, that is a meaningful mispricing.

The reference constraint is the logical floor from Section 1: because a goal is one route to a goal-or-assist, a correctly priced compound contract can never trade below either single-stat leg. A violation of that inequality is a pricing error that needs no model to identify, only three simultaneous quotes.

This is not a riskless arbitrage. The independence approximation carries model risk in both directions: a goal is negatively correlated with the same player's own assist, which pushes true $P(\text{goal} \cap \text{assist})$ below $g a$, while a player heavily involved in attack has elevated odds of both

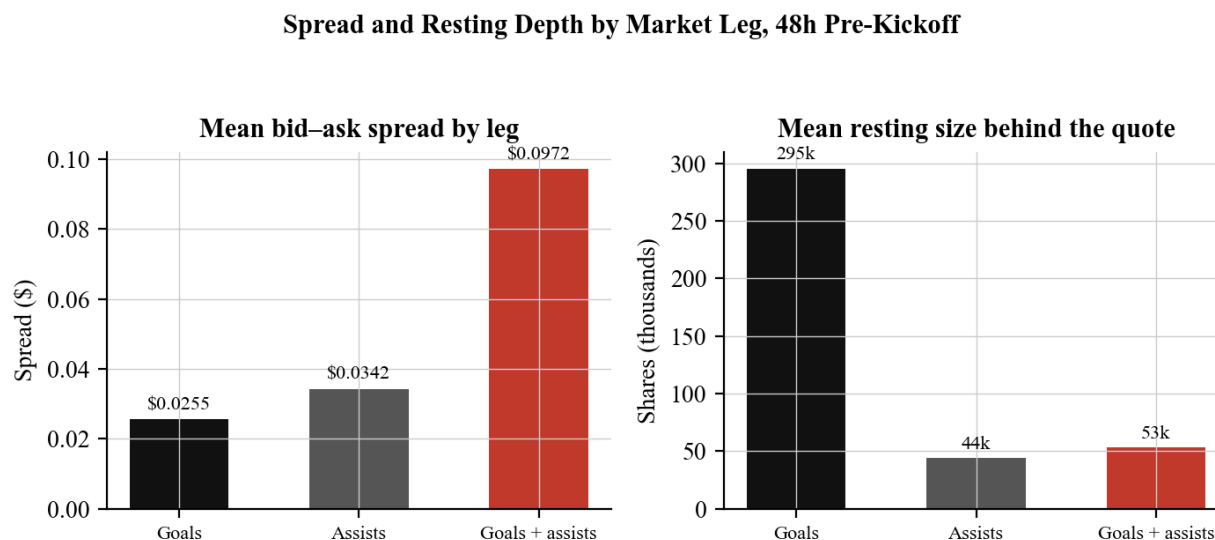
outcomes for reasons outside the formula. The strategy is statistical relative value with adverse-selection exposure: a resting bid improved toward fair value can still be filled by a counterparty who is better informed than the formula.

4. Data Analysis

Every snapshot of quotes and book depth is bucketed by time to kickoff. The four figures below test, against the data, the mechanisms the two strategies depend on.

Spread and Resting Depth by Market Leg

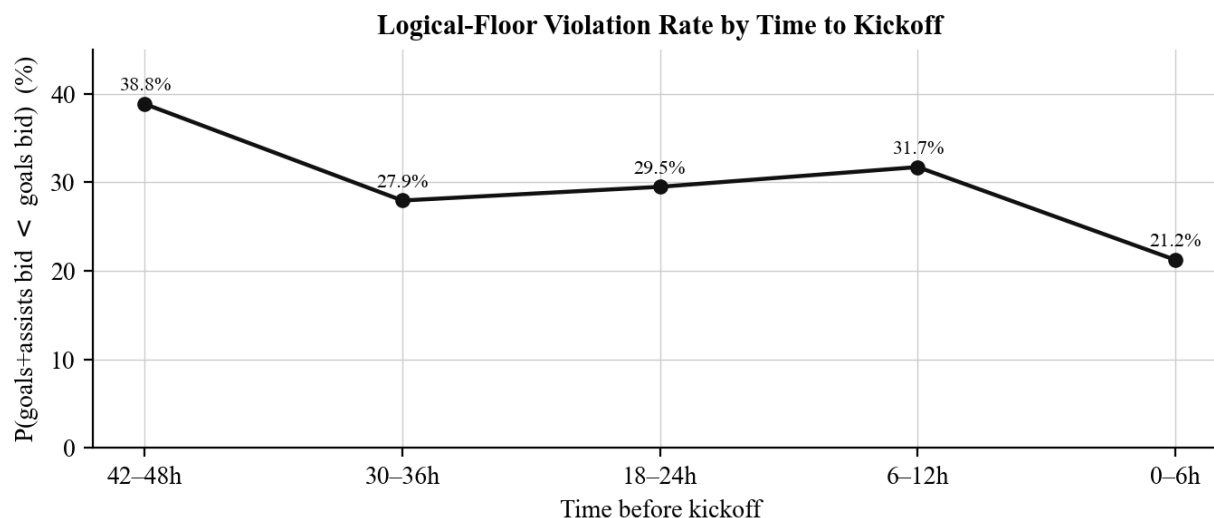
The plot below displays the mean bid–ask spread and the mean resting size for each of the three legs, measured over the forty-eight hours before kickoff for the fourteen highest-coverage players.



The plot shows that the compound contract is the stale one on both measures at once. It carries a $3.81\times$ wider spread than the goals leg (\$0.0972 against \$0.0255) and rests $5.6\times$ less size behind its quotes (52,966 against 295,156 shares), widening to $7.5\times$ in the final four hours. The goals leg is the anchor: its spread stays near \$0.019 for the entire two days, and its book grows from roughly 295,000 to 376,000 shares as retail attention arrives. The compound market behaves nothing like it. It opens at a \$0.157 spread and compresses to \$0.047 by kickoff, but its resting size never grows. Retail flow arrives into the simple market and largely ignores the compound one.

Logical-Floor Violation Rate by Time to Kickoff

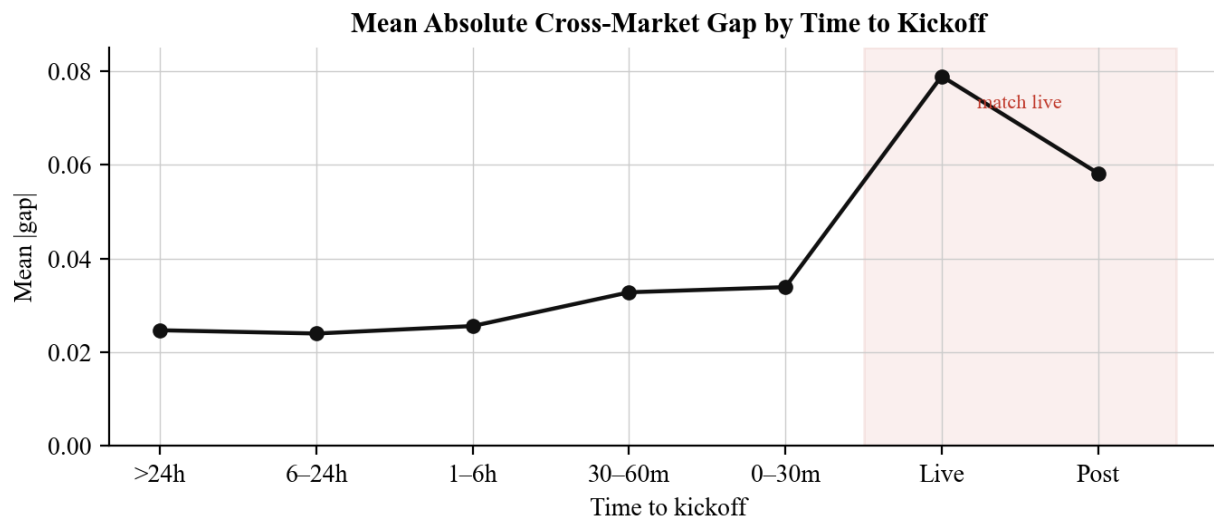
The plot below displays the fraction of pre-kickoff observations in which the compound bid sat below the goals bid, grouped by hours before kickoff.



The plot shows that the best resting bid on the more likely outcome sat below the best resting bid on a subset of it in 28.81% of pre-kickoff observations (35,894 of 124,571), by a mean of \$0.088. The violation rate is highest far from kickoff, 38.83% at forty-two to forty-eight hours out, and decays to 21.23% inside the final six hours, as a de-staling process predicts. It also runs opposite to fame. Messi's compound book violated the identity six times in 7,287 observations, while players one tier below the household names, well known enough for their markets to attract flow but not so well known that the compound leg was watched as closely as the simple one, carried violation rates above 20%.

Mean Absolute Cross-Market Gap by Time to Kickoff

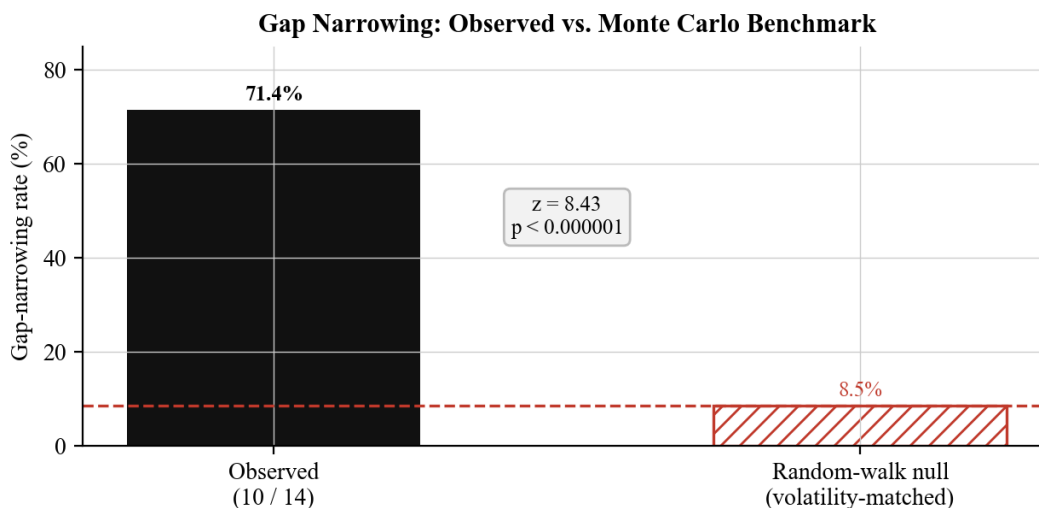
The plot below displays the mean absolute difference between the compound price and its formula fair value, by time bucket, from more than a day before kickoff through live play.



The plot shows that the gap is roughly flat near 0.025 for the entire pre-kickoff window and then more than triples, to 0.079, once the match goes live. Real scoring events hit the three legs at different speeds, and the resulting lag is the inefficiency the second strategy trades.

Gap Narrowing: Observed vs. Monte Carlo Benchmark

To test whether the pre-kickoff mispricing actually reverts rather than merely drifting, per-step increments of *gap* were computed from 157,406 consecutive same-player, same-game observations (median step about sixty seconds), giving a mean of approximately zero and a standard deviation of $\sigma = 0.0248$ per step. For each of the fourteen patience-strategy compound positions with a resolvable gap reading at both entry and exit, 5,000 random-walk paths were simulated, each starting at that position's actual entry gap and taking its actual number of observed steps, and the fraction of paths ending with a smaller absolute gap than they started with was recorded. Exit is the position's own pre-kickoff sale or, where that sale had not filled, kickoff itself, the point at which the strategy is designed to be flat, so every position is scored on the window it was meant to trade whether or not its exit filled in time.



The plot shows that the observed data narrows far more often than the volatility-matched benchmark: 71.4% of the real positions closed toward fair value, against 8.5% of the simulated paths. Because the simulation is calibrated directly from the real minute-by-minute movements and assumes no drift, the difference between 71.4% and 8.5% reflects structure in the price process rather than a parameter choice.

5. Statistical Testing

Each test below is a one-sample proportion z -test. Where the null probability is an entry price rather than a fixed benchmark, the test asks whether the strategy's realized results differ from what the market's own pricing implied. That is the right question for a strategy that might just be fast rather than better informed.

Test 1: Is the gap mean-reversion real?

The null hypothesis is that the gap signal follows a driftless, volatility-matched random walk, which narrows 8.5% of the time.

$$H_0 : p = 0.085 \quad H_1 : p > 0.085 \quad n = 14, \quad \hat{p} = 0.714$$

$$z = \frac{0.714 - 0.085}{\sqrt{\frac{0.085(1 - 0.085)}{14}}} = 8.43, \quad p < 0.000001.$$

The p -value is effectively zero, so we reject the null. At $n = 14$ the normal approximation is loose, so the exact binomial tail is the safer statement: $P(X \geq 10 \mid n = 14, p = 0.085) \approx 1.5 \times 10^{-8}$, the same conclusion. The four positions that fully sold before kickoff, the narrowest possible reading of the strategy, narrowed in three of four. One caveat: contract prices live in $[0, 1]$, so gap is bounded and the simulated walk is not, and a bounded process that starts far from zero has to revert eventually because it has nowhere else to go. Two things limit how much of the result that can explain. First, the effect is large: a boundary artifact does not plausibly produce a 71.4% against 8.5% margin at these gap sizes. Second, the low simulated rate comes from the walk having room to drift over a typical holding period, not from the walk being pushed outward. Re-running the null with reflecting barriers at the price bounds would raise the simulated rate somewhat but would not close the gap.

Test 2: Is the speed strategy calibrated?

The speed strategy trades a confirmed public event before the resting book requotes. Which side it ends up holding depends on the event: a confirmed goal or assist leaves it long “Yes,” a confirmed substitution or full-time whistle leaves it long “No.” Both are the same trade, acting on public news faster than the book adjusts, applied to opposite events, so they are tested together. The null hypothesis is that the market was already pricing those contracts efficiently, so the true win probability equals the average price the account paid for the side it held.

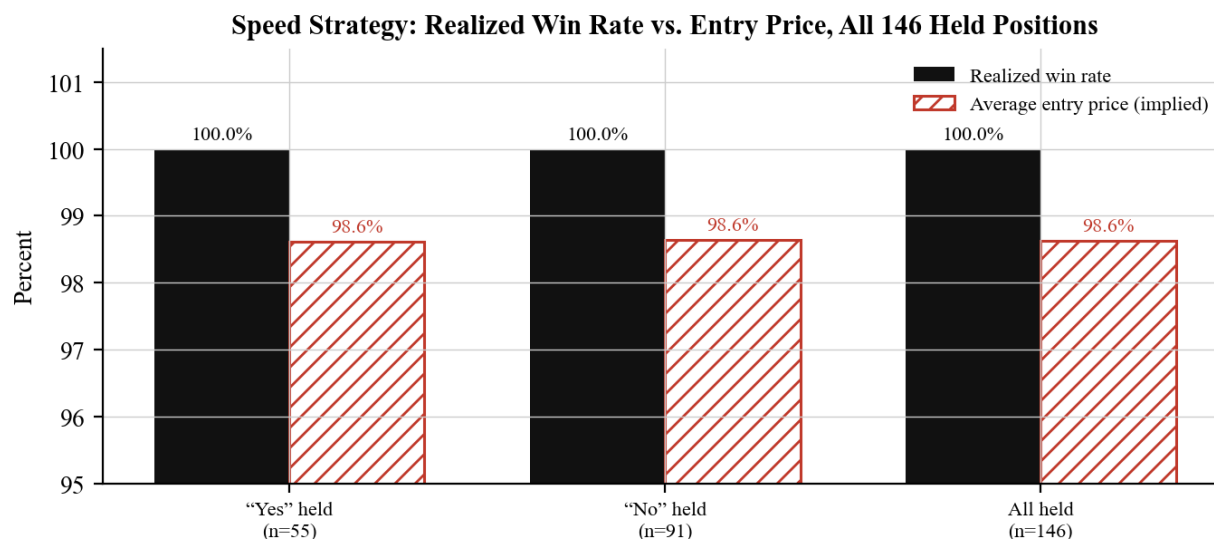
Before the result: this test cannot validate the strategy at these odds. That has to be settled before the outcome is read, because it decides what the outcome can mean. Every held position was bought at \$0.92 to \$0.99, an average of \$0.9863, so the null already expects the strategy to win about 98.6% of the time. With $n = 146$, the best record any strategy can post, winning every single position, gives

$$z_{\max} = \frac{1.000 - 0.9863}{\sqrt{\frac{0.9863(1 - 0.9863)}{146}}} = 1.42, \quad p = 0.155,$$

which does not reach significance. The same holds exactly: under the market's own prices a perfect record has probability $\prod_i p_i \approx 0.13$, well above 0.05. So no outcome this sample could have produced, including a flawless one, would have rejected the null. At these entry prices a perfect record only becomes significant past roughly 217 positions. Whatever the speed strategy's true edge, this test was never able to confirm it here. A failure to reject is therefore the only possible result, and it says nothing about whether the strategy works.

The result.

$$H_0 : p = 0.9863 \quad H_1 : p \neq 0.9863 \quad n = 146, \quad \hat{p} = 1.000, \quad z = 1.42, \quad p = 0.155.$$



The strategy posted that best possible record: all 146 positions carried to a decided settlement resolved the account's way, "Yes" and "No" sides alike, against a 98.6% implied rate. It did not miss once. As established above, the test fails to reject because it cannot do otherwise, not because the strategy fell short.

Two readings remain open, and the data cannot choose between them. The null assumes that the price the account paid, moments after a confirmed event, was already the fair probability. The speed strategy's entire claim is that it was not: that the resting book was still quoting a pre-event price, so a contract bought at \$0.98 was in fact worth close to \$1.00. If that claim is right, the record

reflects a one-to-two-point timing edge. If the null is right, the account should have expected about two losses in 146, and a result at least this profitable would occur about 14% of the time. The profit is real and mechanically reliable. What separates the two readings is more data, not a different test: roughly 217 positions at these prices, or event-level evidence that entries landed below the post-event fair value, which is what the latency sample below measures.

What the win rates do and do not show

The speed strategy's headline win rate, 100% across its 146 decided held positions, invites a reading it cannot support. The strategy buys contracts the market has already priced between roughly 92% and 99% and holds them to settlement, so a near-perfect win rate is what those entry prices predict rather than a result. A perfectly calibrated market handed the same trades would produce nearly the same number. The question that separates edge from arithmetic is whether the realized rate beats the price paid. Test 2 answers it: the realized rate sits above the price paid, but by less than a point and a half, a margin this sample cannot distinguish from chance. That is consistent with a strategy that prices better than the market, and equally consistent with one that does not.

That leaves the speed strategy's profit resting on a timing claim: it buys after a goal, substitution, or whistle is public but before the resting book requotes. A win rate cannot establish this, because a stale price and a fair price at 99% resolve the same way. The only direct evidence is the reaction-time sample in Section 2: seven independently timestamped goals, a mean latency near 91 seconds, every entry landing while the contract still traded below the post-event fact. Seven observations is a real sample but a thin one, and it is the only evidence the speed strategy has. Only the patience strategy has an edge that survives a test against the market's own pricing.

Multiple comparisons and in-sample validity

Bonferroni-correcting the two tests above (threshold $0.05/2 = 0.025$) changes no conclusion: the mean-reversion test clears it by orders of magnitude, and the speed-strategy calibration test does not reach even the uncorrected 0.05. The bigger problem is that the gap signal was designed

and evaluated on the same thirty-five-day dataset, with no out-of-sample holdout. A complete census of every soccer competition Polymarket US lists finds that none of the other seventy leagues currently carries the three-parallel-market structure the signal requires, so a genuine replication needs a second tournament that does not yet exist. The collector already discovers events across all seventy leagues, so that data begins accumulating the moment one lists a player goal or assist market again.

6. Risk

A key risk to the patience strategy arises when the compound contract's staleness does not close before kickoff. The exit is a passive resting sell rather than a market order, so it depends on a counterparty's order arriving. When it does not fill in time, the position is carried into live play, where match variance sits on top of an already-late exit. Of 43 positions, only 13 fully exited before kickoff as intended; the remaining 30 were carried into the match. The 13 clean exits won every position that did not close flat (12 of 12); the 30 delayed positions won only 84.6%, with lower average returns (24.4% per position against 28.5%). That is the cost of fill risk. The clean-exit number is partly a selection artifact: a position moving the wrong way is more likely to still be open at kickoff than sold at a loss beforehand.

Jeremy Doku's goals-or-assists position ahead of Belgium–Senegal is a clear example. It was bought across three pre-kickoff fills at \$0.29 to \$0.31, a blended \$0.2978, and was not sold before kickoff. Carried into the match, it filled across six tranches at \$0.38 to \$0.41, realizing +\$25.18 (+27.6%) and +\$23.57 on its two largest tranches. The mispricing it was entered on was real. It closed late, and a first move the other way would have produced the opposite result on the same delay.

The speed strategy carries a different risk, and it is the mirror image of its win rate. Every held position was entered between \$0.92 and \$0.99, so a winning position earns one to eight cents a share while a position that resolves against the call loses its entire cost. None did: all 146 decided positions resolved the account's way. But the average speed position cost \$318 and returned \$3.92, so a single reversal, a goal overturned on review or a substitution recorded in error, would erase the

profit of roughly eighty winning positions. The strategy's record is only as good as the confirmation behind each entry.

To keep the latency measurement honest, event timestamps are taken only from neutral match reports, never from Polymarket's price data, and second-half goals are excluded rather than approximated. Converting a second-half minute to a wall-clock time requires the real halftime-break length, and assuming a standard fifteen minutes can swing a computed latency by 300 seconds, more than the quantity being measured.

7. Execution Model

Pre-Trade Identification and Filtering

- **Clean three-leg observation.** A candidate surfaces only when all three legs quote from the same pre-kickoff market state and the gap exceeds a threshold; $|gap| > 0.05$ held on 31 of 208 clean samples.
- **Star-player universe.** Positions concentrate on high-coverage names (Messi, Mbappé, Haaland, Kane and similar), whose compound markets de-stale reliably enough to exit into before kickoff. An obscure player can carry the same gap on paper, with far less confidence it closes and far less volume to exit into.

Entry Logic

- **The fair-value ceiling.** Rather than crossing the spread, the patience strategy repeatedly improves its own resting bid in small increments, capped strictly below the formula-implied fair value, so the position buys at or below its own fair-value estimate even in the best case for a counterparty. On the fills, 96.1% of pre-kickoff compound buys with a concurrent quote on both legs (49 of 51) were priced below the formula value computed from those leg asks.
- **Passive only.** Section 4 shows the inefficiency sat almost entirely on the bid side. A crossable, locked arbitrage, buying the compound at its ask and selling the goals leg at its bid, existed in only 0.05% of observations. An aggressive strategy would have found nothing to take.
- **The confirmed-event trigger.** The speed strategy triggers on events, not price levels. On a

confirmed goal, assist, substitution, or final whistle for a tracked player, the near-certain side is bought at whatever price is showing and held to \$1.00 settlement.

Risk and Position Management

- **Fixed-dollar sizing.** Positions are sized as a fixed fraction of available capital rather than a share count, for consistency across contract prices.
- **Hard cap.** Total portfolio risk is capped at 5% of capital, checked at kickoff; a position built past that fraction across multiple fills is cut back with a below-market sell.
- **Breadth over size.** The speed strategy spreads capital across every still-unresolved candidate at the moment of the trigger rather than concentrating on one, since the confirming event makes each outcome close to certain rather than just likely.

Exit Strategy

- **Staged sell-off.** Inside the final five minutes before kickoff, the patience strategy sells roughly a quarter of whatever remains once per minute, crossing the spread each round to guarantee that tranche fills.
- **Unconditional backstop.** A final sweep at kickoff closes anything the schedule leaves behind, so the position is meant to be flat by kickoff, a property the trade record found missing in 30 of 43 cases.
- **Hold to settlement.** Once a speed-strategy entry lands near \$0.98 to \$0.99, the remaining margin is too thin, net of the trading fee, to be worth an active sell. Zero of 43 patience-strategy positions were ever sold at a price at or above \$0.90, and 147 of 149 speed-strategy positions were held to settlement.

The live order-placement layer for the patience strategy was developed jointly with a collaborator on a separate account and is described here as reconstructed from that logic and cross-checked against the surviving trade ledger; every figure in this paper derives from the author's own account and its own records.

8. Final Weights

Two execution schedules follow from the model above. The pre-kickoff exit, which exists to close the fill-risk gap in Section 6, sells a fixed fraction of the remaining position each minute inside the final five, with an unconditional backstop at kickoff:

Minutes to kickoff	Action	Fraction of remainder	Cumulative % closed
5	Staged sell	25%	25.0%
4	Staged sell	25%	43.8%
3	Staged sell	25%	57.8%
2	Staged sell	25%	68.4%
1	Staged sell	25%	76.3%
0	Unconditional sweep	100%	100%

Entries are not scheduled the same way. They are placed as the signal appears, across the full pre-kickoff window. Of 89 pre-kickoff compound buys, the median fill landed 11.7 hours before kickoff (mean 18.6 hours), and the fills weight toward the final six hours rather than spreading evenly:

Window before kickoff	Share of pre-kickoff compound buys
More than 24 hours	21%
18–24 hours	19%
6–18 hours	20%
Final 6 hours	39%

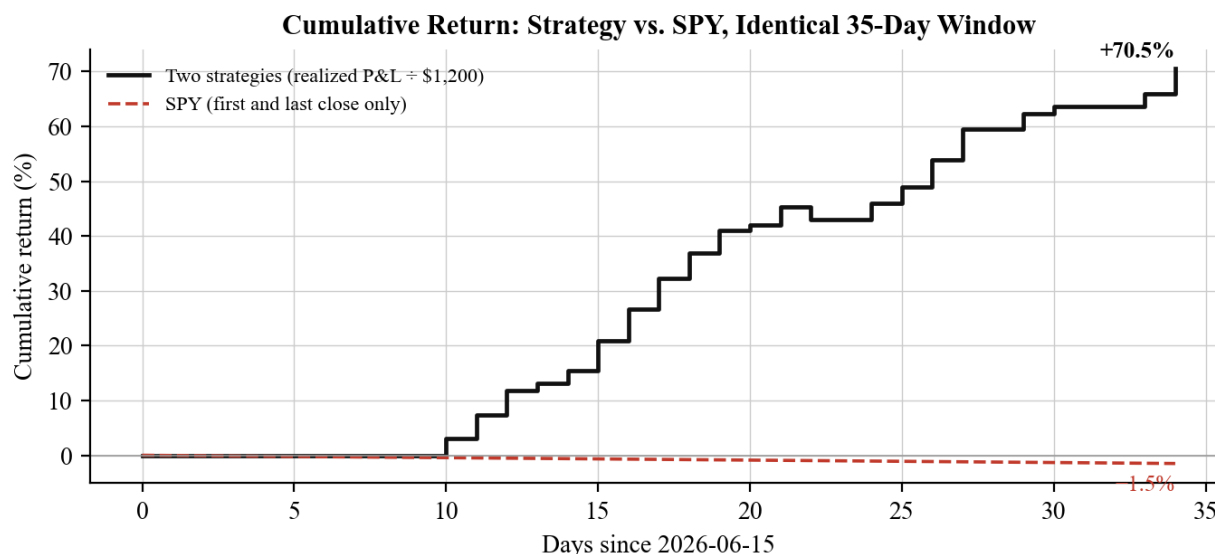
9. Results

The strategies were run against Polymarket US's World Cup player-prop markets over a thirty-five-day window, 2026-06-15 to 2026-07-19. Net capital contributed was \$1,200 across four completed deposits; the account carries zero open positions at close, so every figure reported is fully realized. The two strategies this paper analyzes realized \$845.77 over the window (the account

as a whole, including trades outside both strategies, grew from \$1,200 to \$1,891.58), and that profit has two parts.

\$273.75 came from the patience strategy (43 compound positions, 38 closed by an active sell and 5 carried to settlement) and \$572.02 from the speed strategy (confirmed-event trades held to settlement, \$571.85 on the 147 positions carried to the final whistle, plus \$0.18 on 2 closed beforehand). Only these two strategies are analyzed, and every test and figure below covers them alone.

Against SPY over the same calendar window (real closes \$754.83 on 2026-06-15, \$743.29 on the last trading day before the window ended), the two analyzed strategies returned +70.5% on contributed capital (\$845.77 on \$1,200), against SPY's -1.5%. The comparison establishes only that the return is not an artifact of a broad-market tailwind.



The two strategies behave differently, and only one carries a statistically validated edge:

	Patience (sold)	Speed strategy (held)
Positions	43	147
Win rate	89.5%	100%
Net P&L	+\$273.75	+\$571.85
Validated edge?	Yes ($p < 0.000001$)	Unstable at these odds ($p = 0.155$)

The speed row covers the 147 positions held to settlement (146 decided, one voided), on which the calibration test is run; the strategy realized a further \$0.18 on 2 positions closed beforehand, for a total of \$572.02 as broken out above. Win rates are over positions that did not close flat (34 of 38 for patience, 146 of 146 for speed). The win-rate row is there for completeness but tells you almost nothing about the speed strategy: it bought contracts the market had already priced between 92% and 99%, so a near-perfect win rate is what those entry prices predict rather than a result (Section 5). The “validated edge” row is the one that separates the two. The patience strategy clears it; the speed strategy does not fail it but cannot be tested by it at this sample size, since even its perfect record is not significant at these prices (Section 5).

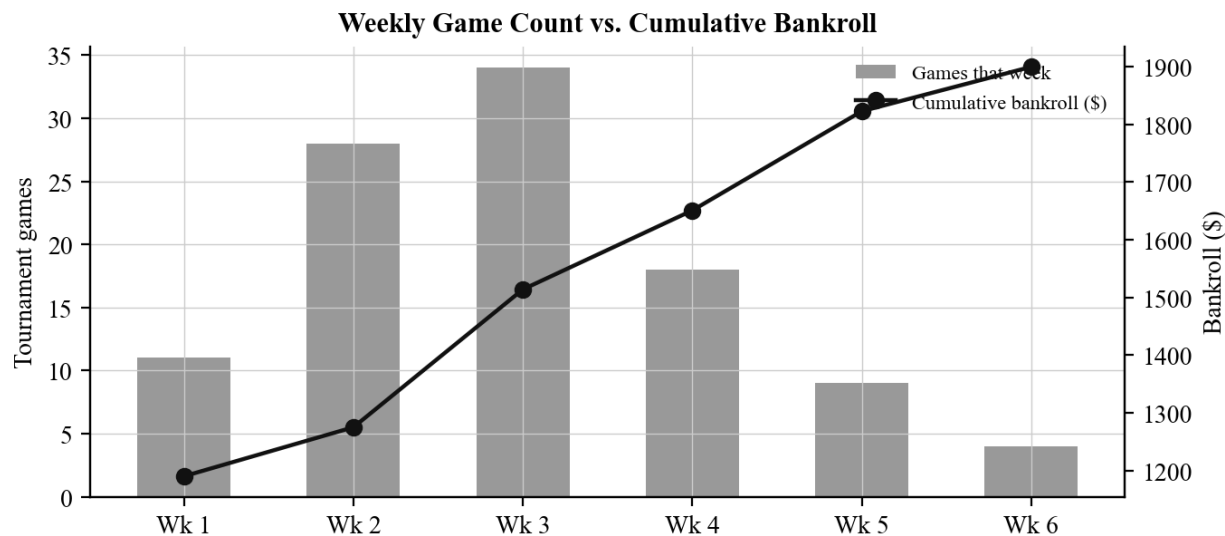
Within the patience strategy, the fill-risk split is the more informative comparison: the 13 positions that exited before kickoff won all 12 that did not close flat, and the 30 carried into live play won 84.6%.

Metric	Strategy	Benchmark (SPY)
Cumulative return, 35 days	+70.5%	−1.5%
Sharpe ratio, daily ($n = 25$)	1.29	n/a
Maximum drawdown	\$27.81 (5.1% of peak)	n/a

The annualized Sharpe is deliberately not reported. Twenty-five daily observations (2026-06-25, the first day a strategy position closed, through 2026-07-19) from a single bounded event put a wide confidence interval on the daily estimate, and scaling that interval by $\sqrt{365}$ produces a range wide enough to be uninformative while still looking precise. Restricting the calculation to the period after all \$1,200 was deposited gives a daily Sharpe of 1.26 on $n = 24$, essentially unchanged once the low-capital ramp-up is removed.

Capacity

The clearest fact in the data about whether this generalizes is that capital grew as opportunity shrank.



Weekly tournament game count ran 11, 28, 34, 18, 9, 4 as the bracket narrowed, while the account's bankroll rose steadily from roughly \$1,190 to \$1,823 across those same weeks. The account held the least capital when opportunity was greatest and the most when it was smallest. With top-of-book depth of 1,406 to 9,762 shares against an average position cost of \$82 (patience) and \$318 (speed), and a median thirty-eight-minute lag between a game-ending event and the settlement that frees the capital, the strategy is capacity-bound by construction.

10. Conclusion

This study demonstrates that a thinly-quoted compound prediction-market contract, priced against the more actively quoted single-stat contracts that decompose it, carries a quantifiable, mechanical pricing inefficiency. Its bid falls below the logical floor $P(\text{goal} \cup \text{assist}) \geq P(\text{goal})$ in 28.81% of 124,571 pre-kickoff observations, and that staleness reverts toward fair value far more often than a volatility-matched random walk predicts: 71.4% against 8.5%, $p < 0.000001$. This is the only edge in the paper that survives a test against the market's own pricing. A second strategy, trading confirmed public events directly, is profitable and never lost a held position, but its results are not evidence of skill: it buys contracts already priced at 92% to 99% and holds them, so its perfect win rate is close to what those prices predict, and a calibration test cannot distinguish it from its own entry pricing (100% realized against 98.6% implied, $p = 0.155$). That is a limit of

the sample, not a failed test: at these prices even a perfect record is not significant until roughly 217 positions, and the result is consistent with a genuine timing edge and with favorable variance alike. What profit it makes rests on a timing claim the win rate cannot test and that only seven directly measured event latencies support. The two strategies converted \$1,200 into \$845.77 in fully realized profit over thirty-five days, \$273.75 from patience and \$572.02 from speed (the account as a whole ended at \$1,891.58), outperforming SPY (+70.5% against -1.5%) over the identical window.

The inefficiency is real, but it is capacity-bound by construction. The account held the least capital when tournament opportunity was greatest and the most when it was smallest, and measured book depth against realized position size leaves no room to scale without redesigning the execution layer. Ultimately, the findings suggest that a retail-heavy prediction market during a marquee event prices its simple contracts efficiently and its compound contracts against itself, an asymmetry a faster, better-modeled counterparty can capture at small size but not, on this evidence, at institutional scale.